

What remains to finish

PetSafe 100-1339 R03 A | v0.9.1 | 8 October 2026

All 150 catalog entries are drawn on one KiCad sheet. This includes empty options and test pads. The drawing is structurally complete; hidden wiring, candidate identities and component values still need evidence.

Drawing / libraries	Electrical gaps	Packages / values
1 A2 sheet; 20 stock + 4 authorized custom symbol definitions; no missing library files for selected candidates.	37 open pads = 18 populated-entry pads + 19 DNP pads. 8 GPIO pads open.	147/150 footprints; 44 capacitor values and 2 magnetic-part values unknown.

What your via review changed

101 IDs reviewed; 26 unmentioned, some photo-matched. V033/V050 rejected, leaving 125 active sites. Latest: V017 joins R42.1/R44.1; R43.2 belongs to V015, not V011-V014; V072-to-R22.2 withdrawn; V073 is GND only. Prior supply, tuning-ground and RC1/VREF corrections remain. Motor VDD stays separate from regulated board VDD.

Remaining area	What is missing	How to resolve
PIC / block interfaces	8 GPIO pads still open: 5, 6, 7, 11, 13, 15, 16, 21. Other pins have local nets but may lack remote continuations. Motor INA/INB origins remain inferred.	Continuity on the numbered PIC map; start with hidden via destinations.
Q8 / RF output / C6	V113/V115 reported GND conflicts with the outer-pad output hypothesis. C6.1 path unresolved despite 10 ohm to P.	Clarify whether GND was measured; compare outer pads to the same GND hole.
Tuning bank	Ground returns corrected. Five local capacitor midpoints are known; connections onward to the antenna are not established.	Check V090/V095/V100/V103/V107 to ANT1 and ANT2.
Auxiliary / receiver	V063 feed unknown. V017 joins R42.1/R44.1; circuit role unknown. V011-V014 and V072 destinations are unresolved after rejected associations.	Identify these remote nets; R44.1 is a photo-selected pad. Receiver bias is still inferred.
D2 / D6 / candidate parts	D2/4P identity and all three pad routes remain open. D6 has two open pads. U6/Q8 and small BJTs/diodes retain candidate identities/polarities.	D2 diode-mode guide; D6 continuity; targeted candidate pin checks.
Other open fitted pads	C26.1 and TP9.1; U6 NC pins 4/5 have unknown external ties. LED common and S1 leg mapping are still provisional.	Targeted continuity; no need to identify chips on empty footprints.
Component values	44 unmarked fitted capacitor values and L1/L2 type/value remain unrecovered. C5 = 470uF / 16V is recorded; 38 fitted resistors have decoded nominal values.	LCR first on RF/tuning parts, recording frequency; in-circuit results are network readings.
Packages	147/150 footprints assigned. C5, S1 and LED1 still need exact geometry; other photo-based package assignments remain candidates.	C5 diameter/lead spacing; S1/LED1 body and pad spacing. R8/C11 sizes already recorded.

Remaining conflicts for later review

Resolved: V025=GND, V026=PIC21/RB0, V073=GND only. R19/V075 remains photo-derived. **V113/V115:** GND reports conflict with the apparent Q8 output connection; those merges remain held. The latest batch adds no resistance or functional measurements.

Validation: KiCad 10.0.5 exports pass; 90 modeled nets match the native file. 55 ERC findings remain: 37 open pins, 10 isolated labels, 5 power and 3 input drive findings. This checks the file, not hardware correctness. No artificial NC or power flags were added.

PIC connections: complete picture

Physical package pins, top view. Use the separate numbered close-up to locate them. OPEN means no modeled pad connection; a local connection can still have an unknown onward destination.

Pin	Function	State	Local destination / evidence
1	RE3/MCLR	Local net	J1 VPP; mapping inferred; D6 route missing
2	RA0	Local net	C25 signal pad / V031; onward destination unknown
3	RA1	Local net	R28 + R29 + TP7; user meter / visible trace
4	RA2	Local net	C39 signal pad / V035; onward destination unknown
5	RA3	OPEN	No visible exit; V036 under-body alignment candidate only
6	RA4	OPEN	No visible exit; V037 under-body alignment candidate only
7	RA5	OPEN	No visible exit; V039 under-body alignment candidate only
8	VSS	Local net	GND / V041 photo anchor
9	RA7/OSC1	Local net	Y1 oscillator; photo-derived
10	RA6/OSC2	Local net	Y1 oscillator; photo-derived
11	RC0	OPEN	No visible exit; V047 under-body alignment candidate only
12	RC1	Local net	VREF board pad / V053; visible local route, remote net unknown
13	RC2	OPEN	No visible exit; V049 under-body alignment candidate only
14	RC3	Local net	TP11; visible route; onward destination unknown
15	RC4	OPEN	V022 user association; remote destination unknown
16	RC5	OPEN	V023 user association; remote destination unknown
17	RC6	Local net	R13; user meter / visible trace
18	RC7	Local net	TP17; visible route; old R13 guess withdrawn
19	VSS	Local net	GND / C32 / V045 photo anchor
20	VDD	Local net	Regulated VDD / C32 / V042 photo anchor
21	RB0	OPEN	V026; user correction. Onward destination unknown
22	RB1	Local net	TP2 / R31; user meter / visible trace
23	RB2	Local net	TP1 / R30; user meter / visible trace
24	RB3	Local net	R4; photo-derived; sensing role remains inferred
25	RB4	Local net	TP4 / C41; user meter / visible trace
26	RB5	Local net	TP10 / R35; user meter / visible trace
27	RB6/CLK	Local net	ICSP CLK / D4 / R32; photo-derived
28	RB7/DAT	Local net	ICSP DAT / D5 / R40; photo-derived

Every remaining open pad

Populated entries (18): C26.1, C6.1, D2.L, D2.R, D2.S, D6.1, D6.2, TP9.1, U4.11, U4.13, U4.15, U4.16, U4.21, U4.5, U4.6, U4.7, U6.L1, U6.L2.

Empty options (19): C42.1, C42.2, C49.1, C49.2, J6.1, J6.2, Q9.S, R36.1, R36.2, R46.1, R46.2, R47.1, R47.2, U6A.L, U6A.R1, U6A.R2, U7.1, U7.2, U7.3.

Pad names above are photo-model names; U6.L1/L2 map to its internal NC pins 4/5. DNP pads do not block understanding of the fitted circuit. Vias with a component association do not automatically prove a connection to every other pad on a previously inferred net.

Evidence: user via review and supplied MX512H diagram; original front/rear photographs; reconstruction.json, via_audit.json, v09_net_changes.json, v091_net_changes.json and native KiCad validation. Pin names follow the existing Microchip datasheet audit. The via viewer retains reports and conflicts.